

# gmanifold: sampling results

## TL;DR

Fit one decoder  $G: \mathbb{R}^{16} \rightarrow \mathbb{R}^D$  per residual location on the real token clouds of SimpleStories-5M, -35M and TinyLlama-1.1B (prefix + every real token; also real story occurrences), then sample on the learned surface within  $\alpha \times$  spacing of the observed states and validate against three independent checks with calibration bands.

- Samples are on-support for  $\alpha \leq 0.7$  at every location of all three models (kernel score above held-out real states, tangent residual and nearest-real distance well inside the noise bands, coverage of the real cloud 0.97–0.99);  $\alpha = 1$  is marginal,  $\alpha = 1.5$  is off-support. Results are stable over seeds, latent dimension and prefixes.
- Pushed through the real model, the samples' images are scored like images of real held-out states by independently fitted destination manifolds.
- The sampler is uniform *within the tube around the observed states* (anchors equalised, clearly not the empirical token density), not in the manifold's own volume; the per-anchor rule that would fill sparse regions collapses.
- The fit is limited by the latent dimension, not by the network or the optimiser (63-fit capacity study); a VAE would be worse for this purpose.
- Two robustness rules matter: drop degenerate and isolated states first, and use the global support radius.
- Retracted negative: the TinyLlama embedding matrix looked "not chartable" (6 spacings) only because five hub tokens with tiny spacing dominate the per-anchor unit; in raw or global-median units it behaves like the small models (details in the last section). The library now detects such hub-dominated clouds.
- To the model itself, samples behave like blends of their nearest real tokens: their next-token distributions are closer to the nearest token's than real tokens are to each other (JS 0.21 vs 0.46 at  $\alpha = 0.3$ ; random-token baseline 0.57), with top-1 agreement 32 % (random 6 %); one-spacing noise loses this (JS 0.51, agreement 8 %). Layer by layer, the images stay nearer to real states than held-out real images do and between the images of their two anchors.

## Summary

**Setup.** GlobalManifold(latent\_dim=16) fitted per residual location on "prefix + every real token" clouds (SimpleStories-5M: 13 locations, 3 seeds,  $m \in \{8, 16, 32\}$ ; SimpleStories-35M: 25 locations, 5 seeds; TinyLlama-1.1B: 8 locations with 2 seeds and all 45 locations with one seed, 31,885 tokens after dropping 111 untrained byte-fallback rows and 1 massive-activation outlier; plus a second prefix for the 5M model). Samples: 2000 per setting from 16,000 candidates, default radius="global".

### What holds everywhere (all models, seeds, prefixes).

1. *On-support regime.* Samples score above held-out real states (kernel  $u \geq 1.03$  at every location, coverage of the real cloud 0.97–0.99) for  $\alpha \leq 0.7$ ; at  $\alpha = 1.0$  they still sit above the half-spacing noise band ( $u$  0.96 vs 0.91) with coverage 0.93;  $\alpha = 1.5$  is off-support ( $u$  0.72–0.74, coverage 0.12). Seed spread of  $u$  at  $\alpha = 0.3$  is  $\leq 0.013$ . The independent tangent-chart residual agrees: samples 0.55–0.64 spacings from the nearest local chart for  $\alpha \leq 0.7$  (held-out real 0.82, half-spacing noise 0.95, one-spacing noise 1.24). The conservative tangent-chart sampler used as a cross-check gives  $u$  0.97 and nearest-real 0.27 at  $\alpha = 0.3$ : on-support but, by construction, hugging the anchors.
2. *Latent dimension.*  $m = 8 / 16 / 32$  give held-out reconstruction 0.78 / 0.74 / 0.67 spacings and  $u$  1.16 / 1.11 / 1.06 at  $\alpha = 0.3$ ; larger  $m$  fits better but the volume weights spread more (ESS 5.5k / 3.4k / 1.5k of 16k).
3. *Volume reweighting works and is validated exactly* on a synthetic sheet with a known volume element (tests/test\_manifold.py): with per-anchor radii the samples reproduce the volume-uniform target; with the global radius they lie between the data density and the target at small  $\alpha$  and converge to the target as  $\alpha$  grows.
4. *Global radius is the robust default.* With per-anchor radii (radius="local") the union of balls is dominated by the sparsest anchors (ball volume  $\propto p^m$ ): on SimpleStories-5M L5. ffn the anchor participation ratio drops from 3546 to  $\sim 100$  and coverage from 0.995 to 0.96; on TinyLlama a single massive-activation state took 100 % of the anchor mass and 99 % of the importance weight (ESS = 1), collapsing every sample onto it. Dropping isolated states (outlier\_mask) and using the global radius removes the failure.
5. *Propagation.* Embedding samples pushed through the real model (embed  $\rightarrow$  L0.attn / L0.ffn / mid / last) are scored by independently fitted destination manifolds like images of real held-out states ( $u$  within  $\pm 0.03$  of the real images at every destination for  $\alpha \leq 0.5$ ,  $\pm 0.2$  at the first sub-layer where the map is near-identity and the samples' own overshoot carries over). Real states displaced by one spacing before propagation are healed by the network:  $u$  0.71 after one sub-layer,  $\geq 0.99$  after six or more.
6. *Loss ablation* (5M, embed and L5. ffn,  $\alpha = 0.3$ ). Removing the local-distance term lowers  $u$  by 0.04–0.07 and raises the samples' reconstruction gap by 25–40 %; 3 $\times$  longer training overfits (train recon 0.51 / 0.40 vs 0.67 / 0.52, held-out recon worse, sample ESS 2.0k / 2.1k vs 4.9k / 6.8k, reconstruction gap  $\times 3$ ). Hidden width and the curvature term are neutral.

7. *Real-occurrence clouds* (20k random token occurrences from 2,500 stories, split by story; all contextual locations of SimpleStories-5M and 35M; §6 of the tables). Lower and token-clustered dimension (TwoNN 5-12), harder fits (held-out reconstruction 0.76-1.17 spacings vs 0.70-0.87 for prefix clouds) and softer calibration bands (one-spacing noise scores  $u$  0.86 instead of 0.69). Samples nevertheless stay on-support at every location of both models for  $\alpha \leq 1$  ( $u \geq 1.01$ , above the  $0.5\times$  band), and their FFN images are scored by the independent destination fits like real held-out images ( $u$  1.01-1.03 vs 1.00). Coverage is lower (0.65 at  $\alpha = 0.3$ ) only because 2000 samples are spread over 18k states.
8. *Decoder capacity study* (§7 of the tables; 63 fits on embed, L2. ffn, L5. ffn). At  $m = 16$  no architecture or training variant changes the held-out reconstruction by more than 0.006 spacings: widths from (256, 128) to (2048, 1024), one to three hidden layers, learning rates  $1e-3$ - $5e-3$ , batches 128-2048,  $\lambda_{\text{geom}}$  0.03-0.3 all give 0.78-0.80 (embed), 0.72-0.74 (L2. ffn), 0.70-0.72 (L5. ffn); the two largest nets are worse (the 3.7M-parameter three-layer net under-fits in 300 epochs), and 600 epochs only lowers the training error (best validation epoch  $\approx 300$ ). The fit is therefore neither capacity- nor optimisation-limited. The one lever is the latent dimension: held-out reconstruction improves smoothly and without plateau from  $m = 8$  to 64 (0.83  $\rightarrow$  0.68, 0.77  $\rightarrow$  0.57, 0.77  $\rightarrow$  0.49), which says the clouds have a long spectral tail rather than a sharp intrinsic dimension (local PCA needs 21-27 components for 90-95 % of the neighbourhood variance). Larger  $m$  costs sampling efficiency: ESS at  $\alpha = 0.3$  drops from 4.9k ( $m = 8$ ) to 25-1200 ( $m = 64$ ) and coverage from 0.99 to 0.81-0.90, while  $u$  moves from 1.15 toward 1.03 (samples closer to the real states' own score).  $m = 16$ -24 is the practical compromise;  $m = 32$  with more candidates if tighter reconstruction matters more than efficiency.
9. *Why not a VAE* (experiments/vae\_check.py, results/vae\_check.json; same architecture with a Gaussian encoder and  $\beta$ -KL, embed and L5. ffn). With  $\beta \leq 1e-3$  the posterior collapses to a point ( $\sigma \approx 0.01$ -0.05), reconstruction equals ours, and decoding prior samples  $z \sim N(0, I)$  fails ( $u$  0.05-0.16, coverage  $\leq 0.15$ ) because the aggregate posterior is nowhere near the prior; our  $\Omega$  sampler on the posterior means works exactly as before. With  $\beta = 1e-2$  the prior sampler becomes usable ( $u$  1.2-1.3, coverage 0.98) at the price of reconstruction (0.79  $\rightarrow$  0.86 at embed, 0.71  $\rightarrow$  0.80 at L5. ffn), of latent dimension (9 of 16 units stay active at embed) and of our sampler's ESS (415 / 126: the KL fights the local-distance term). The prior sampler also targets the aggregate-posterior ( $\approx$  data) density, not manifold volume, and has no notion of "near the observed support". Kept: nothing; noted as useful by-products: the number of active units as a data-driven  $m$  estimate, and the posterior  $\sigma$  as a per-point latent scale.
10. *What "volume-uniform" means on the real embedding cloud* (3940 SimpleStories-5M tokens,  $m = 16$ , 40k samples). Rank correlation between samples-per-real-anchor and the anchor's cell volume ( $\propto r_i^m$ ): default sampler +0.005 ( $\alpha = 0.5$ ) / -0.004 ( $\alpha = 1$ ), data-density sampler -0.275, per-anchor-radius sampler +0.015 with 90 % of its samples on 1 % of the anchors and off-support ( $u$  0.57). So the default sampler is uniform over the uniform-thickness tube  $\Omega$  around the observed states (anchors equalised; clearly not the data density), but not uniform in the manifold's own volume, which would need the large cells of sparse regions to be filled — exactly what the per-anchor rule attempts and fails at. On the synthetic sheet the same effect appears as the global radius sitting between the data density and the exact volume target at small  $\alpha$ .
11. *How the samples look to the model* (§8 of the tables, experiments/downstream.py). With an embedding sample at the last position of the prefix and the full 5M model run: the sample's next-token distribution is closer to that of its nearest real token (JS 0.21 at  $\alpha = 0.3$ , 0.27 at 0.7, 0.37 at 1.5) than real tokens are to their own nearest neighbour (0.46) and far from a random token's (0.57); top-1 predictions agree with the nearest token's in 32 % of the samples (real-to-neighbour 19 %, random 6 %); entropy is slightly higher (3.4 vs 2.9 nats), as expected for a blend. Real tokens displaced by one spacing lose the neighbour relation (JS 0.51, top-1 agreement 8 %, entropy 3.9). Along the network the images of  $\alpha \leq 0.7$  samples stay closer to real states than held-out real images do at every location (0.64-0.91 vs 0.89-0.96 spacings), keep  $u \approx 1$ , and sit 0.5-0.9 spacings from the segment joining the images of their two nearest anchors (real 0.8-0.95, noise 1.1-1.3): they remain interpolations of their neighbours.

**What does not hold, and what we actually tried.** The one cloud that did not behave was the *embedding matrix* of TinyLlama-1.1B (31,885 rows after dropping 111 shared byte-fallback rows and 1 massive-activation token;  $D = 2048$ ). What was run: GlobalManifold with hidden (512, 256), 150 epochs,  $m = 16$  (two seeds) and  $m = 64$  (one seed), 90/10 split by token, the same  $\alpha$  sweep, plus the eight-location and all-45-location sweeps of the same model. What we saw first: held-out reconstruction "6 spacings", samples "2.6 spacings from any real row", TwoNN  $\approx 234$  — and we called the cloud not chartable. What was actually going on: five hub tokens (tight clusters of foreign-script junk tokens such as Архівовано, Webachiv, IABot, spacing  $\approx 0.1$ ) are the nearest neighbour of 74 % of all held-out rows, so "divide by the nearest anchor's spacing" divides by 0.1 for most points. In raw units the decoder reconstructs held-out rows at 0.656 and training rows at 0.646, both  $\approx$  the cloud's nearest-neighbour distance (0.678), the same regime as the SimpleStories embeddings; in units of the global median spacing held-out real rows sit at 0.99 (recon 0.96),  $\alpha = 0.3$  samples at 0.46 (recon 0.09) and one-neighbour-distance noise at 1.40 (recon 1.38). The library now detects hub-dominated clouds (median spacing of the anchors that points actually land on  $<$  half the global median) and reports in global units; the TinyLlama tables were regenerated in those units. What remains true: this cloud is high-dimensional (TwoNN 234 vs 19-24 for SimpleStories, MLE 32) and its coverage by 2000 samples is lower (0.60 vs  $\geq 0.97$  elsewhere); with the corrected unit all four validators behave normally there (held-out real

0.99 / 0.95 / 1.00 / 0.98 on nearest / recon / u / tangent, one-spacing noise 1.40 / 1.39 / 0.68 / 1.40, samples at  $\alpha = 0.3$  0.44 / 0.09 / 1.35 / 0.43). Not tried:  $m > 64$ , more than 150 epochs, frequent-token subsets, other preprocessing. Use `intrinsic_dimension`, the held-out reconstruction and the reported hub share before trusting samples on a new cloud.

Point clouds: a fixed prefix followed by every real token, residual state at the last position, one cloud per residual location; 90 % of the tokens fit the manifold, 10 % are held out. All distances are in units of the local spacing (distance to the 8-th real neighbour);  $u$  is the Guidotti kernel score (1 on fitted states); bands = held-out real states and real states displaced by  $0.5 \times / 1 \times$  their spacing. Samples: 2000 per setting,  $8 \times$  candidates, volume reweighting.

## 0. Runs

| run           | model                              | D    | locations | states | time   |
|---------------|------------------------------------|------|-----------|--------|--------|
| 35M           | SimpleStories/SimpleStories-35M    | 512  | 25        | 3964   | 5 min  |
| 35M_seeds     | SimpleStories/SimpleStories-35M    | 512  | 25        | 3964   | 8 min  |
| 5M            | SimpleStories/SimpleStories-5M     | 256  | 13        | 3940   | 17 min |
| 5M_prefix2    | SimpleStories/SimpleStories-5M     | 256  | 13        | 3940   | 6 min  |
| TinyLlama     | TinyLlama/TinyLlama-1.1B-Chat-v1.0 | 2048 | 8         | 31885  | 13 min |
| TinyLlama_all | TinyLlama/TinyLlama-1.1B-Chat-v1.0 | 2048 | 45        | 31885  | 34 min |
| TinyLlama_m64 | TinyLlama/TinyLlama-1.1B-Chat-v1.0 | 2048 | 5         | 31885  | 9 min  |

## 1. Intrinsic dimension and fit quality per location

**35M** (SimpleStories/SimpleStories-35M)

| location | TwoNN  | MLE    | val recon m=16  |
|----------|--------|--------|-----------------|
| embed    | 23.000 | 28.573 | 0.867 (rank 16) |
| L0.attn  | 23.855 | 28.275 | 0.889 (rank 16) |
| L0.ffn   | 15.243 | 27.655 | 0.801 (rank 16) |
| L1.attn  | 15.390 | 27.768 | 0.806 (rank 16) |
| L1.ffn   | 12.394 | 24.061 | 0.779 (rank 16) |
| L2.attn  | 12.448 | 23.696 | 0.782 (rank 16) |
| L2.ffn   | 11.188 | 20.273 | 0.768 (rank 16) |
| L3.attn  | 10.997 | 20.091 | 0.765 (rank 16) |
| L3.ffn   | 10.393 | 19.375 | 0.756 (rank 16) |
| L4.attn  | 10.332 | 19.209 | 0.753 (rank 16) |
| L4.ffn   | 10.035 | 18.590 | 0.743 (rank 16) |
| L5.attn  | 9.900  | 18.511 | 0.741 (rank 16) |
| L5.ffn   | 9.898  | 17.878 | 0.734 (rank 16) |
| L6.attn  | 9.870  | 17.718 | 0.730 (rank 16) |
| L6.ffn   | 10.133 | 17.067 | 0.724 (rank 16) |
| L7.attn  | 9.956  | 16.577 | 0.721 (rank 16) |
| L7.ffn   | 9.941  | 16.389 | 0.715 (rank 16) |
| L8.attn  | 9.989  | 16.329 | 0.713 (rank 16) |
| L8.ffn   | 9.950  | 16.504 | 0.714 (rank 16) |
| L9.attn  | 9.688  | 16.281 | 0.714 (rank 16) |
| L9.ffn   | 9.844  | 16.523 | 0.715 (rank 16) |
| L10.attn | 9.832  | 16.555 | 0.715 (rank 16) |
| L10.ffn  | 10.124 | 16.779 | 0.717 (rank 16) |
| L11.attn | 10.139 | 16.805 | 0.714 (rank 16) |
| L11.ffn  | 10.741 | 16.916 | 0.713 (rank 16) |

**35M\_seeds** (SimpleStories/SimpleStories-35M)

| location | TwoNN  | MLE    | val recon m=16  |
|----------|--------|--------|-----------------|
| embed    | 23.000 | 28.573 | 0.866 (rank 16) |
| L0.attn  | 23.855 | 28.275 | 0.889 (rank 16) |
| L0.ffn   | 15.243 | 27.655 | 0.804 (rank 16) |
| L1.attn  | 15.390 | 27.768 | 0.806 (rank 16) |
| L1.ffn   | 12.394 | 24.061 | 0.782 (rank 16) |
| L2.attn  | 12.448 | 23.696 | 0.781 (rank 16) |
| L2.ffn   | 11.188 | 20.273 | 0.767 (rank 16) |

| location | TwoNN  | MLE    | val recon m=16  |
|----------|--------|--------|-----------------|
| L3.attn  | 10.997 | 20.091 | 0.767 (rank 16) |
| L3.ffn   | 10.393 | 19.375 | 0.755 (rank 16) |
| L4.attn  | 10.332 | 19.209 | 0.752 (rank 16) |
| L4.ffn   | 10.035 | 18.590 | 0.739 (rank 16) |
| L5.attn  | 9.900  | 18.511 | 0.740 (rank 16) |
| L5.ffn   | 9.898  | 17.878 | 0.734 (rank 16) |
| L6.attn  | 9.870  | 17.718 | 0.729 (rank 16) |
| L6.ffn   | 10.133 | 17.067 | 0.726 (rank 16) |
| L7.attn  | 9.956  | 16.577 | 0.718 (rank 16) |
| L7.ffn   | 9.941  | 16.389 | 0.712 (rank 16) |
| L8.attn  | 9.989  | 16.329 | 0.713 (rank 16) |
| L8.ffn   | 9.950  | 16.504 | 0.715 (rank 16) |
| L9.attn  | 9.688  | 16.281 | 0.711 (rank 16) |
| L9.ffn   | 9.844  | 16.523 | 0.714 (rank 16) |
| L10.attn | 9.832  | 16.555 | 0.713 (rank 16) |
| L10.ffn  | 10.124 | 16.779 | 0.717 (rank 16) |
| L11.attn | 10.139 | 16.805 | 0.715 (rank 16) |
| L11.ffn  | 10.741 | 16.916 | 0.712 (rank 16) |

### 5M (SimpleStories/SimpleStories-5M)

| location | TwoNN  | MLE    | val recon m=8  | val recon m=16  | val recon m=32  |
|----------|--------|--------|----------------|-----------------|-----------------|
| embed    | 18.367 | 21.788 | 0.832 (rank 8) | 0.786 (rank 16) | 0.743 (rank 32) |
| L0.attn  | 18.159 | 21.976 | 0.852 (rank 8) | 0.812 (rank 16) | 0.765 (rank 32) |
| L0.ffn   | 14.250 | 18.821 | 0.783 (rank 8) | 0.750 (rank 16) | 0.703 (rank 32) |
| L1.attn  | 14.342 | 18.790 | 0.783 (rank 8) | 0.747 (rank 16) | 0.703 (rank 32) |
| L1.ffn   | 14.095 | 17.632 | 0.781 (rank 8) | 0.738 (rank 16) | 0.690 (rank 32) |
| L2.attn  | 14.010 | 17.550 | 0.779 (rank 8) | 0.737 (rank 16) | 0.685 (rank 32) |
| L2.ffn   | 13.163 | 16.629 | 0.771 (rank 8) | 0.725 (rank 16) | 0.664 (rank 32) |
| L3.attn  | 13.147 | 16.373 | 0.773 (rank 8) | 0.722 (rank 16) | 0.660 (rank 32) |
| L3.ffn   | 12.594 | 16.125 | 0.768 (rank 8) | 0.714 (rank 16) | 0.646 (rank 32) |
| L4.attn  | 12.470 | 16.183 | 0.765 (rank 8) | 0.711 (rank 16) | 0.636 (rank 32) |
| L4.ffn   | 12.731 | 16.964 | 0.776 (rank 8) | 0.715 (rank 16) | 0.637 (rank 32) |
| L5.attn  | 12.667 | 16.941 | 0.767 (rank 8) | 0.705 (rank 16) | 0.620 (rank 32) |
| L5.ffn   | 14.109 | 17.737 | 0.769 (rank 8) | 0.705 (rank 16) | 0.607 (rank 32) |

### 5M\_prefix2 (SimpleStories/SimpleStories-5M)

| location | TwoNN  | MLE    | val recon m=16  |
|----------|--------|--------|-----------------|
| embed    | 18.367 | 21.788 | 0.785 (rank 16) |
| L0.attn  | 18.433 | 21.762 | 0.798 (rank 16) |
| L0.ffn   | 14.628 | 18.799 | 0.750 (rank 16) |
| L1.attn  | 14.715 | 18.835 | 0.744 (rank 16) |
| L1.ffn   | 14.619 | 17.816 | 0.732 (rank 16) |
| L2.attn  | 14.605 | 17.825 | 0.735 (rank 16) |
| L2.ffn   | 13.523 | 16.819 | 0.723 (rank 16) |
| L3.attn  | 13.298 | 16.477 | 0.717 (rank 16) |
| L3.ffn   | 12.991 | 16.204 | 0.707 (rank 16) |
| L4.attn  | 12.902 | 16.273 | 0.706 (rank 16) |
| L4.ffn   | 13.298 | 16.698 | 0.715 (rank 16) |
| L5.attn  | 13.386 | 16.850 | 0.709 (rank 16) |
| L5.ffn   | 14.251 | 17.093 | 0.705 (rank 16) |

### TinyLlama (TinyLlama/TinyLlama-1.1B-Chat-v1.0)

| location | TwoNN  | MLE    | val recon m=16    |
|----------|--------|--------|-------------------|
| embed    | 234    | 32.065 | 0.953 (rank 16) † |
| L0.attn  | 94.784 | 37.765 | 0.930 (rank 16) † |
| L0.ffn   | 18.079 | 31.489 | 0.976 (rank 16)   |
| L3.ffn   | 11.911 | 26.038 | 0.789 (rank 16)   |

| location | TwoNN  | MLE    | val recon m=16  |
|----------|--------|--------|-----------------|
| L7.ffn   | 10.327 | 22.550 | 0.764 (rank 16) |
| L11.ffn  | 12.867 | 23.396 | 0.770 (rank 16) |
| L15.ffn  | 8.481  | 16.980 | 0.774 (rank 16) |
| L21.ffn  | 9.465  | 18.466 | 0.761 (rank 16) |

† hub-dominated cloud (a few anchors are the nearest neighbour of most points): distances in units of the global median spacing.

#### **TinyLlama\_all** (TinyLlama/TinyLlama-1.1B-Chat-v1.0)

| location | TwoNN  | MLE    | val recon m=16    |
|----------|--------|--------|-------------------|
| embed    | 234    | 32.065 | 0.953 (rank 16) † |
| L0.attn  | 94.784 | 37.765 | 0.930 (rank 16) † |
| L0.ffn   | 18.079 | 31.489 | 0.974 (rank 16)   |
| L1.attn  | 16.461 | 30.534 | 0.915 (rank 16)   |
| L1.ffn   | 14.716 | 27.207 | 0.838 (rank 16)   |
| L2.attn  | 15.036 | 27.849 | 0.825 (rank 16)   |
| L2.ffn   | 15.244 | 30.256 | 0.817 (rank 16)   |
| L3.attn  | 15.355 | 30.845 | 0.805 (rank 16)   |
| L3.ffn   | 11.911 | 26.038 | 0.790 (rank 16)   |
| L4.attn  | 12.464 | 27.062 | 0.786 (rank 16)   |
| L4.ffn   | 11.520 | 26.684 | 0.774 (rank 16)   |
| L5.attn  | 11.702 | 26.673 | 0.772 (rank 16)   |
| L5.ffn   | 11.076 | 25.488 | 0.769 (rank 16)   |
| L6.attn  | 11.231 | 26.003 | 0.766 (rank 16)   |
| L6.ffn   | 10.726 | 24.191 | 0.770 (rank 16)   |
| L7.attn  | 10.723 | 24.274 | 0.770 (rank 16)   |
| L7.ffn   | 10.327 | 22.550 | 0.764 (rank 16)   |
| L8.attn  | 10.442 | 22.710 | 0.763 (rank 16)   |
| L8.ffn   | 10.962 | 23.905 | 0.768 (rank 16)   |
| L9.attn  | 11.348 | 24.256 | 0.766 (rank 16)   |
| L9.ffn   | 11.070 | 23.240 | 0.768 (rank 16)   |
| L10.attn | 11.267 | 23.263 | 0.766 (rank 16)   |
| L10.ffn  | 12.254 | 23.563 | 0.767 (rank 16)   |
| L11.attn | 12.445 | 23.679 | 0.767 (rank 16)   |
| L11.ffn  | 12.867 | 23.396 | 0.770 (rank 16)   |
| L12.attn | 12.944 | 23.486 | 0.769 (rank 16)   |
| L12.ffn  | 11.704 | 21.333 | 0.772 (rank 16)   |
| L13.attn | 11.873 | 21.441 | 0.771 (rank 16)   |
| L13.ffn  | 11.177 | 19.719 | 0.764 (rank 16)   |
| L14.attn | 11.114 | 19.563 | 0.760 (rank 16)   |
| L14.ffn  | 10.010 | 18.826 | 0.761 (rank 16)   |
| L15.attn | 9.900  | 18.706 | 0.762 (rank 16)   |
| L15.ffn  | 8.481  | 16.980 | 0.774 (rank 16)   |
| L16.attn | 8.591  | 17.081 | 0.774 (rank 16)   |
| L16.ffn  | 7.939  | 15.171 | 0.787 (rank 16)   |
| L17.attn | 7.952  | 15.215 | 0.785 (rank 16)   |
| L17.ffn  | 7.916  | 15.268 | 0.786 (rank 16)   |
| L18.attn | 7.962  | 15.259 | 0.784 (rank 16)   |
| L18.ffn  | 8.006  | 15.521 | 0.789 (rank 16)   |
| L19.attn | 8.000  | 15.543 | 0.787 (rank 16)   |
| L19.ffn  | 8.126  | 16.240 | 0.789 (rank 16)   |
| L20.attn | 8.118  | 16.247 | 0.784 (rank 16)   |
| L20.ffn  | 8.395  | 17.297 | 0.785 (rank 16)   |
| L21.attn | 8.449  | 17.337 | 0.781 (rank 16)   |
| L21.ffn  | 9.465  | 18.466 | 0.761 (rank 16)   |

† hub-dominated cloud (a few anchors are the nearest neighbour of most points): distances in units of the global median spacing.

## TinyLlama\_m64 (TinyLlama/TinyLlama-1.1B-Chat-v1.0)

| location | TwoNN  | MLE    | val recon m=64    |
|----------|--------|--------|-------------------|
| embed    | 234    | 32.065 | 0.934 (rank 64) † |
| L0.ffn   | 18.079 | 31.489 | 0.934 (rank 64)   |
| L3.ffn   | 11.911 | 26.038 | 0.730 (rank 64)   |
| L11.ffn  | 12.867 | 23.396 | 0.658 (rank 64)   |
| L21.ffn  | 9.465  | 18.466 | 0.681 (rank 64)   |

† hub-dominated cloud (a few anchors are the nearest neighbour of most points): distances in units of the global median spacing.

## 2. Samples versus $\alpha$ (mean over locations; bands in the last rows)

**35M**, m = 16

| $\alpha$ / band   | nearest real | recon | tangent resid | u (mean) | u (min over loc) | coverage | ESS  | ball multiplicity |
|-------------------|--------------|-------|---------------|----------|------------------|----------|------|-------------------|
| 0.100             | 0.617        | 0.127 | 0.567         | 1.114    | 1.004            | 0.992    | 2724 | 1.088             |
| 0.200             | 0.621        | 0.130 | 0.571         | 1.120    | 1.009            | 0.989    | 2697 | 1.200             |
| 0.300             | 0.622        | 0.136 | 0.568         | 1.117    | 1.095            | 0.985    | 2638 | 1.278             |
| 0.500             | 0.662        | 0.169 | 0.596         | 1.093    | 1.069            | 0.980    | 2479 | 1.337             |
| 0.700             | 0.731        | 0.222 | 0.649         | 1.052    | 1.021            | 0.973    | 2283 | 1.322             |
| 1.000             | 0.868        | 0.344 | 0.761         | 0.958    | 0.922            | 0.929    | 1926 | 1.421             |
| 1.500             | 1.181        | 0.698 | 1.024         | 0.721    | 0.680            | 0.119    | 1288 | 1.866             |
| held-out real     | 0.896        | 0.752 | 0.808         | 0.997    | 0.990            | -        | -    | -                 |
| real + 0.5x noise | 1.022        | 0.909 | 0.945         | 0.905    | 0.901            | -        | -    | -                 |
| real + 1x noise   | 1.330        | 1.270 | 1.265         | 0.674    | 0.666            | -        | -    | -                 |

**35M\_seeds**, m = 16

| $\alpha$ / band   | nearest real | recon | tangent resid | u (mean) | u (min over loc) | coverage | ESS  | ball multiplicity |
|-------------------|--------------|-------|---------------|----------|------------------|----------|------|-------------------|
| 0.100             | 0.616        | 0.128 | 0.566         | 1.114    | 1.006            | 0.992    | 2817 | 1.073             |
| 0.200             | 0.622        | 0.130 | 0.571         | 1.118    | 1.015            | 0.989    | 2790 | 1.172             |
| 0.300             | 0.621        | 0.136 | 0.567         | 1.118    | 1.096            | 0.985    | 2732 | 1.234             |
| 0.500             | 0.660        | 0.168 | 0.595         | 1.093    | 1.071            | 0.980    | 2564 | 1.282             |
| 0.700             | 0.729        | 0.221 | 0.648         | 1.053    | 1.027            | 0.973    | 2338 | 1.310             |
| 1.000             | 0.865        | 0.339 | 0.758         | 0.961    | 0.923            | 0.932    | 1966 | 1.384             |
| 1.500             | 1.175        | 0.685 | 1.019         | 0.726    | 0.683            | 0.122    | 1324 | 1.881             |
| held-out real     | 0.896        | 0.751 | 0.808         | 0.997    | 0.990            | -        | -    | -                 |
| real + 0.5x noise | 1.022        | 0.908 | 0.945         | 0.905    | 0.901            | -        | -    | -                 |
| real + 1x noise   | 1.330        | 1.268 | 1.265         | 0.674    | 0.666            | -        | -    | -                 |

**5M**, m = 8

| $\alpha$ / band   | nearest real | recon | tangent resid | u (mean) | u (min over loc) | coverage | ESS  | ball multiplicity |
|-------------------|--------------|-------|---------------|----------|------------------|----------|------|-------------------|
| 0.100             | 0.656        | 0.092 | 0.605         | 1.160    | 1.139            | 0.995    | 5757 | 1.017             |
| 0.200             | 0.657        | 0.093 | 0.606         | 1.159    | 1.136            | 0.994    | 5661 | 1.034             |
| 0.300             | 0.664        | 0.097 | 0.612         | 1.158    | 1.136            | 0.992    | 5536 | 1.047             |
| 0.500             | 0.683        | 0.108 | 0.625         | 1.151    | 1.129            | 0.988    | 5103 | 1.113             |
| 0.700             | 0.710        | 0.127 | 0.646         | 1.135    | 1.112            | 0.983    | 4222 | 1.244             |
| 1.000             | 0.770        | 0.184 | 0.697         | 1.081    | 1.049            | 0.976    | 2575 | 1.695             |
| 1.500             | 0.949        | 0.354 | 0.853         | 0.919    | 0.866            | 0.950    | 1106 | 3.039             |
| held-out real     | 0.914        | 0.784 | 0.863         | 0.998    | 0.994            | -        | -    | -                 |
| real + 0.5x noise | 1.033        | 0.934 | 0.987         | 0.913    | 0.910            | -        | -    | -                 |
| real + 1x noise   | 1.320        | 1.278 | 1.280         | 0.694    | 0.685            | -        | -    | -                 |

**5M\_prefix2**, m = 16

| $\alpha$ / band | nearest real | recon | tangent resid | u (mean) | u (min over loc) | coverage | ESS  | ball multiplicity |
|-----------------|--------------|-------|---------------|----------|------------------|----------|------|-------------------|
| 0.100           | 0.590        | 0.109 | 0.545         | 1.119    | 1.099            | 0.992    | 3369 | 1.028             |
| 0.200           | 0.598        | 0.114 | 0.550         | 1.116    | 1.096            | 0.989    | 3361 | 1.052             |
| 0.300           | 0.613        | 0.123 | 0.559         | 1.109    | 1.089            | 0.986    | 3335 | 1.065             |
| 0.500           | 0.664        | 0.154 | 0.592         | 1.086    | 1.067            | 0.976    | 3212 | 1.094             |

| $\alpha$ / band   | nearest real | recon | tangent resid | u (mean) | u (min over loc) | coverage | ESS  | ball multiplicity |
|-------------------|--------------|-------|---------------|----------|------------------|----------|------|-------------------|
| 0.700             | 0.735        | 0.204 | 0.643         | 1.049    | 1.029            | 0.967    | 3013 | 1.124             |
| 1.000             | 0.876        | 0.320 | 0.749         | 0.962    | 0.938            | 0.932    | 2462 | 1.258             |
| 1.500             | 1.198        | 0.667 | 1.010         | 0.730    | 0.683            | 0.111    | 1374 | 1.914             |
| held-out real     | 0.920        | 0.733 | 0.823         | 0.998    | 0.993            | -        | -    | -                 |
| real + 0.5x noise | 1.038        | 0.888 | 0.951         | 0.913    | 0.910            | -        | -    | -                 |
| real + 1x noise   | 1.330        | 1.241 | 1.251         | 0.695    | 0.686            | -        | -    | -                 |

#### TinyLlama, m = 16

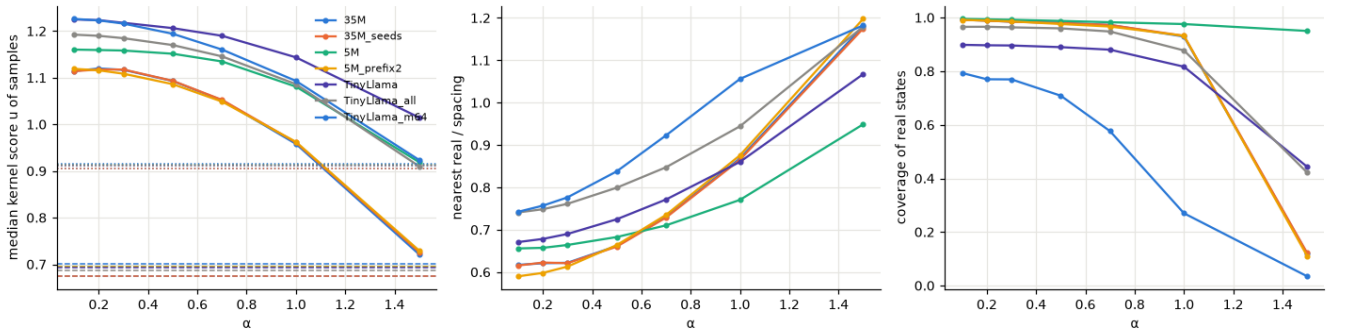
| $\alpha$ / band   | nearest real | recon | tangent resid | u (mean) | u (min over loc) | coverage | ESS | ball multiplicity |
|-------------------|--------------|-------|---------------|----------|------------------|----------|-----|-------------------|
| 0.100             | 0.671        | 0.080 | 0.588         | 1.225    | 1.135            | 0.898    | 789 | 1.000             |
| 0.200             | 0.678        | 0.085 | 0.595         | 1.223    | 1.131            | 0.897    | 796 | 1.000             |
| 0.300             | 0.690        | 0.092 | 0.606         | 1.218    | 1.125            | 0.896    | 806 | 1.001             |
| 0.500             | 0.725        | 0.114 | 0.642         | 1.206    | 1.110            | 0.890    | 849 | 1.016             |
| 0.700             | 0.771        | 0.144 | 0.687         | 1.190    | 1.085            | 0.880    | 888 | 1.066             |
| 1.000             | 0.860        | 0.205 | 0.771         | 1.144    | 1.005            | 0.817    | 835 | 1.281             |
| 1.500             | 1.067        | 0.374 | 0.968         | 1.015    | 0.764            | 0.443    | 497 | 2.075             |
| held-out real     | 0.920        | 0.840 | 0.865         | 0.998    | 0.996            | -        | -   | -                 |
| real + 0.5x noise | 1.048        | 0.981 | 0.999         | 0.912    | 0.904            | -        | -   | -                 |
| real + 1x noise   | 1.361        | 1.315 | 1.321         | 0.693    | 0.675            | -        | -   | -                 |

#### TinyLlama\_all, m = 16

| $\alpha$ / band   | nearest real | recon | tangent resid | u (mean) | u (min over loc) | coverage | ESS  | ball multiplicity |
|-------------------|--------------|-------|---------------|----------|------------------|----------|------|-------------------|
| 0.100             | 0.741        | 0.076 | 0.633         | 1.192    | 1.139            | 0.966    | 762  | 1.000             |
| 0.200             | 0.748        | 0.081 | 0.641         | 1.190    | 1.134            | 0.966    | 778  | 1.000             |
| 0.300             | 0.761        | 0.088 | 0.655         | 1.185    | 1.131            | 0.964    | 807  | 1.002             |
| 0.500             | 0.799        | 0.108 | 0.694         | 1.170    | 1.112            | 0.960    | 901  | 1.018             |
| 0.700             | 0.847        | 0.135 | 0.744         | 1.146    | 1.086            | 0.948    | 1025 | 1.077             |
| 1.000             | 0.944        | 0.191 | 0.837         | 1.086    | 1.024            | 0.878    | 1085 | 1.315             |
| 1.500             | 1.179        | 0.348 | 1.058         | 0.910    | 0.778            | 0.422    | 654  | 2.267             |
| held-out real     | 0.879        | 0.793 | 0.806         | 0.999    | 0.996            | -        | -    | -                 |
| real + 0.5x noise | 1.011        | 0.939 | 0.947         | 0.911    | 0.904            | -        | -    | -                 |
| real + 1x noise   | 1.329        | 1.281 | 1.280         | 0.687    | 0.666            | -        | -    | -                 |

#### TinyLlama\_m64, m = 64

| $\alpha$ / band   | nearest real | recon | tangent resid | u (mean) | u (min over loc) | coverage | ESS  | ball multiplicity |
|-------------------|--------------|-------|---------------|----------|------------------|----------|------|-------------------|
| 0.100             | 0.742        | 0.062 | 0.639         | 1.226    | 1.114            | 0.793    | 1718 | 1.000             |
| 0.200             | 0.757        | 0.066 | 0.653         | 1.223    | 1.108            | 0.770    | 1667 | 1.000             |
| 0.300             | 0.776        | 0.070 | 0.668         | 1.216    | 1.094            | 0.769    | 1596 | 1.000             |
| 0.500             | 0.838        | 0.088 | 0.719         | 1.194    | 1.059            | 0.709    | 1419 | 1.000             |
| 0.700             | 0.923        | 0.111 | 0.788         | 1.160    | 1.011            | 0.576    | 1234 | 1.001             |
| 1.000             | 1.056        | 0.156 | 0.894         | 1.093    | 0.916            | 0.270    | 991  | 1.002             |
| 1.500             | 1.182        | 0.254 | 0.986         | 0.924    | 0.667            | 0.035    | 646  | 1.057             |
| held-out real     | 0.938        | 0.788 | 0.824         | 0.998    | 0.996            | -        | -    | -                 |
| real + 0.5x noise | 1.064        | 0.935 | 0.962         | 0.915    | 0.905            | -        | -    | -                 |
| real + 1x noise   | 1.376        | 1.277 | 1.289         | 0.700    | 0.678            | -        | -    | -                 |



Dotted / dashed lines: kernel score of real states displaced by 0.5x / 1x spacing (the off-support bands).

### 3. Latent dimension and seed sensitivity (35M)

| m  | $\alpha$ | u mean | u range (loc $\times$ seed) | nearest real | coverage | ESS  | val recon range |
|----|----------|--------|-----------------------------|--------------|----------|------|-----------------|
| 16 | 0.300    | 1.117  | 1.095-1.205                 | 0.622        | 0.985    | 2638 | 0.711-0.892     |
| 16 | 0.500    | 1.093  | 1.069-1.191                 | 0.662        | 0.980    | 2479 | 0.711-0.892     |
| 16 | 1.000    | 0.958  | 0.922-1.080                 | 0.868        | 0.929    | 1926 | 0.711-0.892     |

### 3. Latent dimension and seed sensitivity (35M\_seeds)

| m  | $\alpha$ | u mean | u range (loc $\times$ seed) | nearest real | coverage | ESS  | val recon range |
|----|----------|--------|-----------------------------|--------------|----------|------|-----------------|
| 16 | 0.300    | 1.118  | 1.096-1.189                 | 0.621        | 0.985    | 2732 | 0.709-0.895     |
| 16 | 0.500    | 1.093  | 1.071-1.183                 | 0.660        | 0.980    | 2564 | 0.709-0.895     |
| 16 | 1.000    | 0.961  | 0.923-1.081                 | 0.865        | 0.932    | 1966 | 0.709-0.895     |

### 3. Latent dimension and seed sensitivity (5M)

| m  | $\alpha$ | u mean | u range (loc $\times$ seed) | nearest real | coverage | ESS  | val recon range |
|----|----------|--------|-----------------------------|--------------|----------|------|-----------------|
| 8  | 0.300    | 1.158  | 1.136-1.217                 | 0.664        | 0.992    | 5536 | 0.763-0.856     |
| 8  | 0.500    | 1.151  | 1.129-1.209                 | 0.683        | 0.988    | 5103 | 0.763-0.856     |
| 8  | 1.000    | 1.081  | 1.049-1.152                 | 0.770        | 0.976    | 2575 | 0.763-0.856     |
| 16 | 0.300    | 1.110  | 1.094-1.172                 | 0.614        | 0.988    | 3436 | 0.704-0.816     |
| 16 | 0.500    | 1.087  | 1.071-1.152                 | 0.662        | 0.979    | 3336 | 0.704-0.816     |
| 16 | 1.000    | 0.965  | 0.944-1.025                 | 0.867        | 0.944    | 2574 | 0.704-0.816     |
| 32 | 0.300    | 1.058  | 1.039-1.110                 | 0.535        | 0.958    | 1490 | 0.607-0.768     |
| 32 | 0.500    | 1.010  | 0.993-1.061                 | 0.628        | 0.927    | 1138 | 0.607-0.768     |
| 32 | 1.000    | 0.786  | 0.760-0.827                 | 1.001        | 0.241    | 569  | 0.607-0.768     |

### 3. Latent dimension and seed sensitivity (5M\_prefix2)

| m  | $\alpha$ | u mean | u range (loc $\times$ seed) | nearest real | coverage | ESS  | val recon range |
|----|----------|--------|-----------------------------|--------------|----------|------|-----------------|
| 16 | 0.300    | 1.109  | 1.089-1.168                 | 0.613        | 0.986    | 3335 | 0.703-0.800     |
| 16 | 0.500    | 1.086  | 1.067-1.138                 | 0.664        | 0.976    | 3212 | 0.703-0.800     |
| 16 | 1.000    | 0.962  | 0.938-1.015                 | 0.876        | 0.932    | 2462 | 0.703-0.800     |

### 3. Latent dimension and seed sensitivity (TinyLlama)

| m  | $\alpha$ | u mean | u range (loc $\times$ seed) | nearest real | coverage | ESS | val recon range |
|----|----------|--------|-----------------------------|--------------|----------|-----|-----------------|
| 16 | 0.300    | 1.218  | 1.125-1.375                 | 0.690        | 0.896    | 806 | 0.761-0.978     |
| 16 | 0.500    | 1.206  | 1.110-1.388                 | 0.725        | 0.890    | 849 | 0.761-0.978     |
| 16 | 1.000    | 1.144  | 1.005-1.418                 | 0.860        | 0.817    | 835 | 0.761-0.978     |

### 4. Sampler variants and loss ablation (5M, $\alpha = 0.3$ )

Sampler variants (same decoder): support radius rule (global =  $\alpha \times \text{median } \rho$  for all anchors, local =  $\alpha \times \rho_i$ ), anchor probability  $\propto R^p$  and whether the volume reweighting is applied. anchors = participation ratio of the anchor distribution.

| location | variant                            | anchors | ESS   | coverage | nearest real | recon | u     |
|----------|------------------------------------|---------|-------|----------|--------------|-------|-------|
| embed    | tangent charts (cross-check)       | -       | 2000  | 0.738    | 0.274        | 0.805 | 0.966 |
| embed    | radius=global, p=m, reweight=True  | 3546    | 4898  | 0.991    | 0.710        | 0.107 | 1.215 |
| embed    | radius=global, p=0, reweight=False | 3546    | 16000 | 0.992    | 0.775        | 0.069 | 1.241 |
| embed    | radius=local, p=m, reweight=True   | 336     | 8555  | 0.960    | 0.668        | 0.134 | 1.193 |
| embed    | radius=local, p=0, reweight=True   | 3546    | 432   | 0.958    | 0.666        | 0.132 | 1.183 |
| embed    | radius=local, p=0, reweight=False  | 3546    | 16000 | 0.997    | 0.774        | 0.068 | 1.240 |
| L2.ffn   | tangent charts (cross-check)       | -       | 2000  | 0.866    | 0.274        | 0.770 | 0.971 |
| L2.ffn   | radius=global, p=m, reweight=True  | 3546    | 6063  | 0.991    | 0.663        | 0.093 | 1.148 |
| L2.ffn   | radius=global, p=0, reweight=False | 3546    | 16000 | 0.992    | 0.719        | 0.064 | 1.218 |
| L2.ffn   | radius=local, p=m, reweight=True   | 755     | 7868  | 0.970    | 0.626        | 0.115 | 1.126 |
| L2.ffn   | radius=local, p=0, reweight=True   | 3546    | 1033  | 0.970    | 0.626        | 0.113 | 1.127 |
| L2.ffn   | radius=local, p=0, reweight=False  | 3546    | 16000 | 0.995    | 0.720        | 0.064 | 1.218 |

| location | variant                            | anchors | ESS    | coverage | nearest real | recon | u     |
|----------|------------------------------------|---------|--------|----------|--------------|-------|-------|
| L5.ffn   | tangent charts (cross-check)       | -       | 2000   | 0.836    | 0.274        | 0.756 | 0.975 |
| L5.ffn   | radius=global, p=m, reweight=True  | 3546    | 6780   | 0.995    | 0.694        | 0.073 | 1.165 |
| L5.ffn   | radius=global, p=0, reweight=False | 3546    | 16000  | 0.995    | 0.725        | 0.052 | 1.201 |
| L5.ffn   | radius=local, p=m, reweight=True   | 104     | 4807   | 0.964    | 0.384        | 0.126 | 1.055 |
| L5.ffn   | radius=local, p=0, reweight=True   | 3546    | 40.125 | 0.973    | 0.437        | 0.153 | 1.090 |
| L5.ffn   | radius=local, p=0, reweight=False  | 3546    | 16000  | 0.996    | 0.724        | 0.052 | 1.201 |

Loss / architecture variants (each refitted, then sampled at  $\alpha = 0.3$ ).

| location | variant     | val recon | train recon | ESS  | coverage | nearest real | recon | u     |
|----------|-------------|-----------|-------------|------|----------|--------------|-------|-------|
| embed    | default     | 0.833     | 0.670       | 4898 | 0.991    | 0.710        | 0.107 | 1.215 |
| embed    | no_geom     | 0.843     | 0.578       | 4374 | 0.983    | 0.612        | 0.133 | 1.150 |
| embed    | no_curv     | 0.832     | 0.670       | 4868 | 0.990    | 0.704        | 0.106 | 1.216 |
| embed    | strong_geom | 0.844     | 0.707       | 5880 | 0.992    | 0.780        | 0.078 | 1.237 |
| embed    | wide        | 0.827     | 0.655       | 4027 | 0.990    | 0.690        | 0.108 | 1.209 |
| embed    | epochs_900  | 0.894     | 0.511       | 1967 | 0.960    | 0.575        | 0.345 | 1.112 |
| L2.ffn   | default     | 0.772     | 0.582       | 6063 | 0.991    | 0.663        | 0.093 | 1.148 |
| L2.ffn   | no_geom     | 0.781     | 0.501       | 4899 | 0.991    | 0.578        | 0.137 | 1.104 |
| L2.ffn   | no_curv     | 0.771     | 0.582       | 6083 | 0.992    | 0.662        | 0.090 | 1.149 |
| L2.ffn   | strong_geom | 0.779     | 0.612       | 5422 | 0.992    | 0.682        | 0.075 | 1.160 |
| L2.ffn   | wide        | 0.772     | 0.565       | 6077 | 0.993    | 0.640        | 0.095 | 1.141 |
| L2.ffn   | epochs_900  | 0.821     | 0.435       | 2320 | 0.972    | 0.558        | 0.350 | 1.067 |
| L5.ffn   | default     | 0.767     | 0.523       | 6780 | 0.995    | 0.694        | 0.073 | 1.165 |
| L5.ffn   | no_geom     | 0.778     | 0.459       | 7407 | 0.997    | 0.630        | 0.101 | 1.126 |
| L5.ffn   | no_curv     | 0.768     | 0.523       | 6802 | 0.993    | 0.690        | 0.072 | 1.166 |
| L5.ffn   | strong_geom | 0.778     | 0.544       | 6167 | 0.992    | 0.705        | 0.063 | 1.180 |
| L5.ffn   | wide        | 0.771     | 0.507       | 6003 | 0.995    | 0.675        | 0.080 | 1.157 |
| L5.ffn   | epochs_900  | 0.813     | 0.397       | 2078 | 0.988    | 0.551        | 0.273 | 1.083 |

## 5. Propagation through the real model: images vs independent destination fits

Samples at the embedding are pushed through the real Transformer stretch embed  $\rightarrow$  dst (fixed prefix context) and scored by the destination manifold fitted only on real destination states. Controls: images of held-out real states, and of real states displaced by  $0.5\times / 1\times$  spacing *before* propagation.

### 35M: embed $\rightarrow$ L0.attn

| set                      | nearest real | recon | tangent resid | u     |
|--------------------------|--------------|-------|---------------|-------|
| sample $\alpha=0.25$     | 0.833        | 0.425 | 0.793         | 1.173 |
| sample $\alpha=0.5$      | 0.729        | 0.420 | 0.690         | 1.173 |
| sample $\alpha=1.0$      | 0.880        | 0.649 | 0.822         | 1.056 |
| real held-out images     | 1.028        | 0.892 | 0.981         | 0.990 |
| real + 0.5x noise images | 1.140        | 1.046 | 1.103         | 0.905 |
| real + 1x noise images   | 1.484        | 1.427 | 1.436         | 0.703 |
| band: held-out real      | 1.028        | 0.892 | 0.981         | 0.990 |
| band: real + 0.5x noise  | 1.144        | 1.041 | 1.104         | 0.909 |
| band: real + 1x noise    | 1.452        | 1.409 | 1.410         | 0.716 |

### 35M: embed $\rightarrow$ L0.ffn

| set                      | nearest real | recon | tangent resid | u     |
|--------------------------|--------------|-------|---------------|-------|
| sample $\alpha=0.25$     | 0.843        | 0.814 | 0.759         | 1.009 |
| sample $\alpha=0.5$      | 0.756        | 0.662 | 0.682         | 1.046 |
| sample $\alpha=1.0$      | 0.820        | 0.653 | 0.749         | 1.070 |
| real held-out images     | 0.937        | 0.802 | 0.878         | 0.996 |
| real + 0.5x noise images | 0.968        | 0.829 | 0.909         | 1.014 |
| real + 1x noise images   | 2.279        | 1.901 | 2.198         | 0.994 |
| band: held-out real      | 0.937        | 0.802 | 0.878         | 0.996 |
| band: real + 0.5x noise  | 1.055        | 0.968 | 1.004         | 0.917 |

| set                   | nearest real | recon | tangent resid | u     |
|-----------------------|--------------|-------|---------------|-------|
| band: real + 1x noise | 1.362        | 1.307 | 1.320         | 0.677 |

### 35M: embed → L5.ffn

| set                      | nearest real | recon | tangent resid | u     |
|--------------------------|--------------|-------|---------------|-------|
| sample $\alpha=0.25$     | 0.837        | 0.656 | 0.741         | 1.012 |
| sample $\alpha=0.5$      | 0.745        | 0.594 | 0.646         | 1.039 |
| sample $\alpha=1.0$      | 0.802        | 0.579 | 0.699         | 1.080 |
| real held-out images     | 0.867        | 0.734 | 0.784         | 0.996 |
| real + 0.5x noise images | 0.877        | 0.707 | 0.789         | 1.028 |
| real + 1x noise images   | 1.006        | 0.734 | 0.967         | 1.060 |
| band: held-out real      | 0.867        | 0.734 | 0.784         | 0.996 |
| band: real + 0.5x noise  | 1.001        | 0.894 | 0.926         | 0.904 |
| band: real + 1x noise    | 1.320        | 1.257 | 1.244         | 0.670 |

### 35M: embed → L11.ffn

| set                      | nearest real | recon | tangent resid | u     |
|--------------------------|--------------|-------|---------------|-------|
| sample $\alpha=0.25$     | 0.882        | 0.670 | 0.752         | 1.003 |
| sample $\alpha=0.5$      | 0.805        | 0.629 | 0.682         | 1.009 |
| sample $\alpha=1.0$      | 0.863        | 0.628 | 0.745         | 1.018 |
| real held-out images     | 0.900        | 0.714 | 0.790         | 0.998 |
| real + 0.5x noise images | 0.917        | 0.702 | 0.811         | 1.005 |
| real + 1x noise images   | 0.988        | 0.728 | 0.918         | 1.017 |
| band: held-out real      | 0.900        | 0.714 | 0.790         | 0.998 |
| band: real + 0.5x noise  | 1.022        | 0.872 | 0.924         | 0.904 |
| band: real + 1x noise    | 1.332        | 1.237 | 1.249         | 0.669 |

### 5M: embed → L0.attn

| set                      | nearest real | recon | tangent resid | u     |
|--------------------------|--------------|-------|---------------|-------|
| sample $\alpha=0.25$     | 0.724        | 0.231 | 0.667         | 1.202 |
| sample $\alpha=0.5$      | 0.738        | 0.253 | 0.682         | 1.196 |
| sample $\alpha=1.0$      | 0.791        | 0.351 | 0.726         | 1.129 |
| real held-out images     | 0.965        | 0.856 | 0.904         | 0.994 |
| real + 0.5x noise images | 1.090        | 1.003 | 1.033         | 0.911 |
| real + 1x noise images   | 1.396        | 1.361 | 1.358         | 0.709 |
| band: held-out real      | 0.965        | 0.856 | 0.904         | 0.994 |
| band: real + 0.5x noise  | 1.087        | 1.003 | 1.030         | 0.914 |
| band: real + 1x noise    | 1.379        | 1.341 | 1.338         | 0.721 |

### 5M: embed → L0.ffn

| set                      | nearest real | recon | tangent resid | u     |
|--------------------------|--------------|-------|---------------|-------|
| sample $\alpha=0.25$     | 0.835        | 0.734 | 0.772         | 1.033 |
| sample $\alpha=0.5$      | 0.841        | 0.722 | 0.777         | 1.038 |
| sample $\alpha=1.0$      | 0.855        | 0.718 | 0.781         | 1.030 |
| real held-out images     | 0.902        | 0.783 | 0.856         | 1.001 |
| real + 0.5x noise images | 1.005        | 0.904 | 0.962         | 0.945 |
| real + 1x noise images   | 1.246        | 1.170 | 1.213         | 0.783 |
| band: held-out real      | 0.902        | 0.783 | 0.856         | 1.001 |
| band: real + 0.5x noise  | 1.026        | 0.940 | 0.986         | 0.914 |
| band: real + 1x noise    | 1.320        | 1.281 | 1.283         | 0.695 |

### 5M: embed → L2.ffn

| set                      | nearest real | recon | tangent resid | u     |
|--------------------------|--------------|-------|---------------|-------|
| sample $\alpha=0.25$     | 0.900        | 0.751 | 0.824         | 0.996 |
| sample $\alpha=0.5$      | 0.896        | 0.746 | 0.822         | 0.997 |
| sample $\alpha=1.0$      | 0.899        | 0.750 | 0.820         | 0.994 |
| real held-out images     | 0.895        | 0.772 | 0.851         | 0.998 |
| real + 0.5x noise images | 0.986        | 0.845 | 0.942         | 0.982 |

| set                     | nearest real | recon | tangent resid | u     |
|-------------------------|--------------|-------|---------------|-------|
| real + 1x noise images  | 1.172        | 1.039 | 1.130         | 0.896 |
| band: held-out real     | 0.895        | 0.772 | 0.851         | 0.998 |
| band: real + 0.5x noise | 1.018        | 0.924 | 0.972         | 0.912 |
| band: real + 1x noise   | 1.309        | 1.266 | 1.268         | 0.691 |

#### 5M: embed → L5.ffn

| set                      | nearest real | recon | tangent resid | u     |
|--------------------------|--------------|-------|---------------|-------|
| sample $\alpha=0.25$     | 0.912        | 0.761 | 0.837         | 0.991 |
| sample $\alpha=0.5$      | 0.915        | 0.763 | 0.842         | 0.992 |
| sample $\alpha=1.0$      | 0.922        | 0.769 | 0.848         | 0.993 |
| real held-out images     | 0.920        | 0.767 | 0.858         | 0.998 |
| real + 0.5x noise images | 0.982        | 0.819 | 0.935         | 0.999 |
| real + 1x noise images   | 1.057        | 0.875 | 1.017         | 0.995 |
| band: held-out real      | 0.920        | 0.767 | 0.858         | 0.998 |
| band: real + 0.5x noise  | 1.031        | 0.916 | 0.981         | 0.910 |
| band: real + 1x noise    | 1.325        | 1.260 | 1.273         | 0.697 |

#### 5M\_prefix2: embed → L0.attn

| set                      | nearest real | recon | tangent resid | u     |
|--------------------------|--------------|-------|---------------|-------|
| sample $\alpha=0.25$     | 0.651        | 0.354 | 0.607         | 1.141 |
| sample $\alpha=0.5$      | 0.691        | 0.414 | 0.634         | 1.113 |
| sample $\alpha=1.0$      | 0.866        | 0.654 | 0.774         | 0.992 |
| real held-out images     | 0.953        | 0.800 | 0.871         | 0.993 |
| real + 0.5x noise images | 1.083        | 0.949 | 1.004         | 0.914 |
| real + 1x noise images   | 1.361        | 1.302 | 1.294         | 0.719 |
| band: held-out real      | 0.953        | 0.800 | 0.871         | 0.993 |
| band: real + 0.5x noise  | 1.081        | 0.953 | 0.998         | 0.914 |
| band: real + 1x noise    | 1.366        | 1.301 | 1.296         | 0.712 |

#### 5M\_prefix2: embed → L0.ffn

| set                      | nearest real | recon | tangent resid | u     |
|--------------------------|--------------|-------|---------------|-------|
| sample $\alpha=0.25$     | 0.737        | 0.598 | 0.669         | 1.037 |
| sample $\alpha=0.5$      | 0.762        | 0.623 | 0.688         | 1.026 |
| sample $\alpha=1.0$      | 0.900        | 0.761 | 0.805         | 0.960 |
| real held-out images     | 0.907        | 0.751 | 0.828         | 1.000 |
| real + 0.5x noise images | 1.009        | 0.867 | 0.940         | 0.947 |
| real + 1x noise images   | 1.260        | 1.147 | 1.190         | 0.790 |
| band: held-out real      | 0.907        | 0.751 | 0.828         | 1.000 |
| band: real + 0.5x noise  | 1.030        | 0.906 | 0.948         | 0.914 |
| band: real + 1x noise    | 1.322        | 1.256 | 1.258         | 0.696 |

#### 5M\_prefix2: embed → L2.ffn

| set                      | nearest real | recon | tangent resid | u     |
|--------------------------|--------------|-------|---------------|-------|
| sample $\alpha=0.25$     | 0.807        | 0.668 | 0.711         | 1.000 |
| sample $\alpha=0.5$      | 0.826        | 0.676 | 0.726         | 0.997 |
| sample $\alpha=1.0$      | 0.921        | 0.744 | 0.818         | 0.977 |
| real held-out images     | 0.917        | 0.724 | 0.815         | 0.998 |
| real + 0.5x noise images | 0.990        | 0.803 | 0.908         | 0.979 |
| real + 1x noise images   | 1.165        | 0.999 | 1.101         | 0.897 |
| band: held-out real      | 0.917        | 0.724 | 0.815         | 0.998 |
| band: real + 0.5x noise  | 1.029        | 0.876 | 0.944         | 0.914 |
| band: real + 1x noise    | 1.321        | 1.224 | 1.241         | 0.695 |

#### 5M\_prefix2: embed → L5.ffn

| set                  | nearest real | recon | tangent resid | u     |
|----------------------|--------------|-------|---------------|-------|
| sample $\alpha=0.25$ | 0.857        | 0.662 | 0.715         | 0.995 |
| sample $\alpha=0.5$  | 0.873        | 0.671 | 0.744         | 0.995 |

| set                      | nearest real | recon | tangent resid | u     |
|--------------------------|--------------|-------|---------------|-------|
| sample $\alpha=1.0$      | 0.942        | 0.729 | 0.818         | 0.988 |
| real held-out images     | 0.928        | 0.708 | 0.815         | 0.996 |
| real + 0.5x noise images | 0.984        | 0.746 | 0.878         | 0.999 |
| real + 1x noise images   | 1.064        | 0.826 | 0.974         | 0.993 |
| band: held-out real      | 0.928        | 0.708 | 0.815         | 0.996 |
| band: real + 0.5x noise  | 1.042        | 0.861 | 0.942         | 0.912 |
| band: real + 1x noise    | 1.339        | 1.225 | 1.246         | 0.693 |

#### TinyLlama: embed $\rightarrow$ L0.attn

| set                      | nearest real | recon | tangent resid | u     |
|--------------------------|--------------|-------|---------------|-------|
| sample $\alpha=0.25$     | 0.691        | 0.582 | 0.681         | 1.158 |
| sample $\alpha=0.5$      | 0.705        | 0.595 | 0.694         | 1.160 |
| sample $\alpha=1.0$      | 0.758        | 0.637 | 0.741         | 1.191 |
| real held-out images     | 0.978        | 0.930 | 0.969         | 0.996 |
| real + 0.5x noise images | 1.090        | 1.052 | 1.081         | 0.901 |
| real + 1x noise images   | 1.376        | 1.351 | 1.367         | 0.674 |
| band: held-out real      | 0.978        | 0.930 | 0.969         | 0.996 |
| band: real + 0.5x noise  | 1.099        | 1.058 | 1.089         | 0.904 |
| band: real + 1x noise    | 1.398        | 1.369 | 1.388         | 0.677 |

#### TinyLlama: embed $\rightarrow$ L0.ffn

| set                      | nearest real | recon | tangent resid | u     |
|--------------------------|--------------|-------|---------------|-------|
| sample $\alpha=0.25$     | 1.807        | 1.465 | 1.706         | 1.011 |
| sample $\alpha=0.5$      | 1.778        | 1.417 | 1.670         | 1.018 |
| sample $\alpha=1.0$      | 1.828        | 1.483 | 1.715         | 1.038 |
| real held-out images     | 1.053        | 0.974 | 1.016         | 0.998 |
| real + 0.5x noise images | 1.281        | 1.183 | 1.232         | 0.954 |
| real + 1x noise images   | 2.772        | 2.603 | 2.710         | 0.795 |
| band: held-out real      | 1.053        | 0.974 | 1.016         | 0.998 |
| band: real + 0.5x noise  | 1.170        | 1.107 | 1.138         | 0.940 |
| band: real + 1x noise    | 1.484        | 1.431 | 1.446         | 0.773 |

#### TinyLlama: embed $\rightarrow$ L7.ffn

| set                      | nearest real | recon | tangent resid | u     |
|--------------------------|--------------|-------|---------------|-------|
| sample $\alpha=0.25$     | 0.974        | 0.735 | 0.885         | 1.008 |
| sample $\alpha=0.5$      | 0.968        | 0.735 | 0.877         | 1.007 |
| sample $\alpha=1.0$      | 0.980        | 0.741 | 0.883         | 1.002 |
| real held-out images     | 0.858        | 0.764 | 0.798         | 0.999 |
| real + 0.5x noise images | 0.891        | 0.761 | 0.833         | 1.006 |
| real + 1x noise images   | 0.985        | 0.780 | 0.938         | 1.020 |
| band: held-out real      | 0.858        | 0.764 | 0.798         | 0.999 |
| band: real + 0.5x noise  | 0.992        | 0.914 | 0.938         | 0.907 |
| band: real + 1x noise    | 1.313        | 1.262 | 1.272         | 0.675 |

#### TinyLlama: embed $\rightarrow$ L21.ffn

| set                      | nearest real | recon | tangent resid | u     |
|--------------------------|--------------|-------|---------------|-------|
| sample $\alpha=0.25$     | 0.877        | 0.688 | 0.737         | 1.012 |
| sample $\alpha=0.5$      | 0.879        | 0.686 | 0.744         | 1.014 |
| sample $\alpha=1.0$      | 0.887        | 0.678 | 0.752         | 1.014 |
| real held-out images     | 0.864        | 0.761 | 0.755         | 0.998 |
| real + 0.5x noise images | 0.880        | 0.763 | 0.778         | 1.000 |
| real + 1x noise images   | 0.935        | 0.758 | 0.838         | 1.007 |
| band: held-out real      | 0.864        | 0.761 | 0.755         | 0.998 |
| band: real + 0.5x noise  | 0.997        | 0.913 | 0.904         | 0.910 |
| band: real + 1x noise    | 1.318        | 1.261 | 1.247         | 0.689 |

## 6. Real-occurrence clouds (random token occurrences in real stories, split by story)

Same fit and sampling protocol on residual states of real token occurrences (20k occurrences, 8 per story,  $\leq 256$  tokens per story); duplicates and isolated outliers dropped from the training split. Propagation uses the tokenwise FFN maps  $L_k.\text{attn} \rightarrow L_k.\text{ffn}$ .

**35M\_stories** (SimpleStories/SimpleStories-35M: 20000 occurrences from 2500 stories, 2000 held out)

| location | train states | TwoNN  | MLE    | val recon |
|----------|--------------|--------|--------|-----------|
| L0.attn  | 17169        | 8.641  | 2.014  | 1.039     |
| L0.ffn   | 17134        | 8.388  | 1.946  | 1.267     |
| L1.attn  | 17933        | 5.659  | 4.501  | 0.942     |
| L1.ffn   | 17933        | 5.545  | 4.348  | 0.991     |
| L2.attn  | 17933        | 7.024  | 6.105  | 0.890     |
| L2.ffn   | 17933        | 7.030  | 5.949  | 0.915     |
| L3.attn  | 17933        | 8.436  | 7.844  | 0.853     |
| L3.ffn   | 17933        | 8.377  | 7.453  | 0.865     |
| L4.attn  | 17933        | 9.793  | 8.837  | 0.839     |
| L4.ffn   | 17933        | 9.604  | 8.384  | 0.840     |
| L5.attn  | 17933        | 10.431 | 9.824  | 0.821     |
| L5.ffn   | 17933        | 10.267 | 9.454  | 0.826     |
| L6.attn  | 17933        | 10.588 | 10.323 | 0.804     |
| L6.ffn   | 17933        | 10.330 | 10.024 | 0.805     |
| L7.attn  | 17933        | 11.219 | 11.512 | 0.792     |
| L7.ffn   | 17933        | 11.080 | 11.327 | 0.792     |
| L8.attn  | 17933        | 11.423 | 12.275 | 0.782     |
| L8.ffn   | 17933        | 11.279 | 12.195 | 0.783     |
| L9.attn  | 17933        | 11.951 | 13.551 | 0.781     |
| L9.ffn   | 17933        | 11.991 | 13.623 | 0.779     |
| L10.attn | 17933        | 12.795 | 15.012 | 0.772     |
| L10.ffn  | 17933        | 12.912 | 15.344 | 0.775     |
| L11.attn | 17933        | 13.301 | 16.178 | 0.769     |
| L11.ffn  | 17933        | 13.743 | 17.434 | 0.764     |

| $\alpha$ / band   | nearest real | recon | u (mean) | u (min over loc) | coverage | ESS  |
|-------------------|--------------|-------|----------|------------------|----------|------|
| 0.100             | 0.888        | 0.238 | 1.120    | 1.012            | 0.784    | 2484 |
| 0.300             | 0.908        | 0.251 | 1.115    | 1.012            | 0.777    | 2477 |
| 0.500             | 0.949        | 0.272 | 1.107    | 1.012            | 0.757    | 2418 |
| 0.700             | 1.006        | 0.305 | 1.094    | 1.010            | 0.720    | 2254 |
| 1.000             | 1.104        | 0.361 | 1.059    | 1.007            | 0.578    | 1754 |
| 1.500             | 1.311        | 0.496 | 0.967    | 0.875            | 0.214    | 996  |
| held-out real     | 0.809        | 0.854 | 0.998    | 0.996            | -        | -    |
| real + 0.5x noise | 0.949        | 0.998 | 0.942    | 0.912            | -        | -    |
| real + 1x noise   | 1.277        | 1.342 | 0.794    | 0.703            | -        | -    |

L0.attn  $\rightarrow$  L0.ffn

| set                     | nearest real | recon | u     |
|-------------------------|--------------|-------|-------|
| sample $\alpha=0.3$     | 1.575        | 2.821 | 1.003 |
| sample $\alpha=0.7$     | 1.888        | 3.024 | 1.004 |
| sample $\alpha=1.0$     | 2.036        | 3.098 | 1.003 |
| real held-out images    | 0.829        | 1.267 | 1.000 |
| real + 1x noise images  | 1.258        | 1.767 | 0.999 |
| band: held-out real     | 0.829        | 1.267 | 1.000 |
| band: real + 0.5x noise | 0.967        | 1.392 | 0.996 |
| band: real + 1x noise   | 1.280        | 1.719 | 0.989 |

L1.attn  $\rightarrow$  L1.ffn

| set                 | nearest real | recon | u     |
|---------------------|--------------|-------|-------|
| sample $\alpha=0.3$ | 0.942        | 0.854 | 1.056 |

| set                     | nearest real | recon | u     |
|-------------------------|--------------|-------|-------|
| sample $\alpha=0.7$     | 1.004        | 0.920 | 1.063 |
| sample $\alpha=1.0$     | 1.075        | 1.013 | 1.058 |
| real held-out images    | 0.699        | 0.991 | 0.999 |
| real + 1x noise images  | 1.138        | 1.362 | 0.960 |
| band: held-out real     | 0.699        | 0.991 | 0.999 |
| band: real + 0.5x noise | 0.858        | 1.124 | 0.978 |
| band: real + 1x noise   | 1.209        | 1.453 | 0.916 |

L2.attn  $\rightarrow$  L2.ffn

| set                     | nearest real | recon | u     |
|-------------------------|--------------|-------|-------|
| sample $\alpha=0.3$     | 0.910        | 0.701 | 1.064 |
| sample $\alpha=0.7$     | 0.972        | 0.800 | 1.053 |
| sample $\alpha=1.0$     | 1.066        | 0.915 | 1.039 |
| real held-out images    | 0.739        | 0.915 | 0.999 |
| real + 1x noise images  | 1.165        | 1.286 | 0.927 |
| band: held-out real     | 0.739        | 0.915 | 0.999 |
| band: real + 0.5x noise | 0.891        | 1.052 | 0.964 |
| band: real + 1x noise   | 1.238        | 1.382 | 0.863 |

L3.attn  $\rightarrow$  L3.ffn

| set                     | nearest real | recon | u     |
|-------------------------|--------------|-------|-------|
| sample $\alpha=0.3$     | 0.832        | 0.598 | 1.058 |
| sample $\alpha=0.7$     | 0.919        | 0.718 | 1.047 |
| sample $\alpha=1.0$     | 1.030        | 0.844 | 1.030 |
| real held-out images    | 0.777        | 0.865 | 0.998 |
| real + 1x noise images  | 1.173        | 1.231 | 0.893 |
| band: held-out real     | 0.777        | 0.865 | 0.998 |
| band: real + 0.5x noise | 0.923        | 1.004 | 0.951 |
| band: real + 1x noise   | 1.258        | 1.344 | 0.818 |

**5M\_stories** (SimpleStories/SimpleStories-5M: 20000 occurrences from 2500 stories, 2000 held out)

| location | train states | TwoNN  | MLE    | val recon |
|----------|--------------|--------|--------|-----------|
| L0.attn  | 16877        | 8.327  | 2.005  | 1.167     |
| L0.ffn   | 16788        | 8.199  | 2.081  | 1.225     |
| L1.attn  | 17933        | 5.294  | 3.782  | 0.899     |
| L1.ffn   | 17933        | 5.291  | 3.999  | 0.914     |
| L2.attn  | 17933        | 6.312  | 5.015  | 0.858     |
| L2.ffn   | 17933        | 6.302  | 5.195  | 0.852     |
| L3.attn  | 17933        | 7.109  | 6.569  | 0.809     |
| L3.ffn   | 17933        | 7.152  | 6.813  | 0.800     |
| L4.attn  | 17933        | 9.227  | 9.133  | 0.773     |
| L4.ffn   | 17933        | 9.333  | 9.464  | 0.771     |
| L5.attn  | 17933        | 11.614 | 12.317 | 0.760     |
| L5.ffn   | 17933        | 12.209 | 13.917 | 0.763     |

| $\alpha$ / band   | nearest real | recon | u (mean) | u (min over loc) | coverage | ESS  |
|-------------------|--------------|-------|----------|------------------|----------|------|
| 0.100             | 0.897        | 0.293 | 1.081    | 1.016            | 0.676    | 3014 |
| 0.300             | 0.918        | 0.302 | 1.079    | 1.014            | 0.652    | 2991 |
| 0.500             | 0.955        | 0.316 | 1.073    | 1.015            | 0.608    | 2830 |
| 0.700             | 1.006        | 0.340 | 1.065    | 1.015            | 0.546    | 2556 |
| 1.000             | 1.088        | 0.377 | 1.043    | 1.015            | 0.425    | 2023 |
| 1.500             | 1.261        | 0.465 | 0.978    | 0.887            | 0.204    | 1281 |
| held-out real     | 0.760        | 0.883 | 0.998    | 0.997            | -        | -    |
| real + 0.5x noise | 0.905        | 1.031 | 0.963    | 0.929            | -        | -    |
| real + 1x noise   | 1.239        | 1.387 | 0.862    | 0.749            | -        | -    |

L0.attn  $\rightarrow$  L0.ffn

| set                     | nearest real | recon | u     |
|-------------------------|--------------|-------|-------|
| sample $\alpha=0.3$     | 1.542        | 2.166 | 1.009 |
| sample $\alpha=0.7$     | 1.672        | 2.317 | 1.010 |
| sample $\alpha=1.0$     | 1.719        | 2.376 | 1.010 |
| real held-out images    | 0.788        | 1.225 | 0.999 |
| real + 1x noise images  | 1.245        | 1.709 | 0.990 |
| band: held-out real     | 0.787        | 1.225 | 0.999 |
| band: real + 0.5x noise | 0.927        | 1.368 | 0.995 |
| band: real + 1x noise   | 1.254        | 1.747 | 0.988 |

L1.attn → L1.ffn

| set                     | nearest real | recon | u     |
|-------------------------|--------------|-------|-------|
| sample $\alpha=0.3$     | 0.885        | 0.811 | 1.033 |
| sample $\alpha=0.7$     | 0.944        | 0.879 | 1.029 |
| sample $\alpha=1.0$     | 0.994        | 0.942 | 1.021 |
| real held-out images    | 0.665        | 0.914 | 0.999 |
| real + 1x noise images  | 1.153        | 1.340 | 0.952 |
| band: held-out real     | 0.665        | 0.914 | 0.999 |
| band: real + 0.5x noise | 0.825        | 1.058 | 0.978 |
| band: real + 1x noise   | 1.186        | 1.399 | 0.917 |

L2.attn → L2.ffn

| set                     | nearest real | recon | u     |
|-------------------------|--------------|-------|-------|
| sample $\alpha=0.3$     | 0.847        | 0.682 | 1.030 |
| sample $\alpha=0.7$     | 0.895        | 0.745 | 1.023 |
| sample $\alpha=1.0$     | 0.968        | 0.813 | 1.017 |
| real held-out images    | 0.712        | 0.852 | 0.999 |
| real + 1x noise images  | 1.165        | 1.258 | 0.935 |
| band: held-out real     | 0.712        | 0.852 | 0.999 |
| band: real + 0.5x noise | 0.861        | 0.996 | 0.966 |
| band: real + 1x noise   | 1.211        | 1.333 | 0.876 |

L3.attn → L3.ffn

| set                     | nearest real | recon | u     |
|-------------------------|--------------|-------|-------|
| sample $\alpha=0.3$     | 0.775        | 0.532 | 1.038 |
| sample $\alpha=0.7$     | 0.841        | 0.595 | 1.032 |
| sample $\alpha=1.0$     | 0.912        | 0.643 | 1.019 |
| real held-out images    | 0.748        | 0.800 | 0.999 |
| real + 1x noise images  | 1.162        | 1.171 | 0.914 |
| band: held-out real     | 0.748        | 0.800 | 0.999 |
| band: real + 0.5x noise | 0.895        | 0.946 | 0.956 |
| band: real + 1x noise   | 1.236        | 1.292 | 0.831 |

## 7. Decoder capacity and hyper-parameters (SimpleStories-5M, three locations)

Every setting scored by held-out reconstruction / spacing (final and best epoch on the validation split) and by the samples at  $\alpha = 0.3$ . default = hidden (512, 256), 300 epochs, lr 2e-3, batch 512,  $\lambda_{\text{geom}}$  0.1.

**embed** — latent dimension (default architecture):

| m  | val recon (best) | train recon | u     | nearest real | coverage | ESS    |
|----|------------------|-------------|-------|--------------|----------|--------|
| 8  | 0.831            | 0.670       | 1.215 | 0.710        | 0.991    | 4898   |
| 16 | 0.788            | 0.563       | 1.149 | 0.642        | 0.986    | 1907   |
| 24 | 0.763            | 0.479       | 1.121 | 0.579        | 0.955    | 946    |
| 32 | 0.743            | 0.423       | 1.093 | 0.552        | 0.933    | 876    |
| 48 | 0.707            | 0.355       | 1.072 | 0.504        | 0.872    | 361    |
| 64 | 0.676            | 0.307       | 1.050 | 0.454        | 0.823    | 24.636 |

**embed** — architecture / training variants at m = 16 (sorted by held-out reconstruction):

| variant                                         | params | val recon | best (epoch) | train recon | u     | coverage | ESS  |
|-------------------------------------------------|--------|-----------|--------------|-------------|-------|----------|------|
| hidden (512, 512, 256), 300 ep                  | 1.06M  | 0.782     | 0.782 @ 299  | 0.506       | 1.120 | 0.955    | 1859 |
| hidden (512, 256), 300 ep, {'lam_geom': 0.03}   | 0.53M  | 0.786     | 0.786 @ 275  | 0.555       | 1.144 | 0.984    | 1554 |
| hidden (512, 256), 300 ep, {'lam_geom': 0.3}    | 0.53M  | 0.789     | 0.788 @ 250  | 0.572       | 1.157 | 0.984    | 2676 |
| default                                         | 0.53M  | 0.788     | 0.788 @ 275  | 0.563       | 1.149 | 0.986    | 1907 |
| hidden (2048, 1024), 300 ep                     | 5.28M  | 0.790     | 0.790 @ 299  | 0.550       | 1.152 | 0.977    | 1468 |
| hidden (512, 256), 600 ep                       | 0.53M  | 0.799     | 0.792 @ 325  | 0.448       | 1.087 | 0.946    | 1240 |
| hidden (512, 256), 300 ep, {'lr': 0.001}        | 0.53M  | 0.792     | 0.792 @ 299  | 0.593       | 1.168 | 0.987    | 2918 |
| hidden (512, 256), 300 ep, {'lr': 0.005}        | 0.53M  | 0.792     | 0.792 @ 299  | 0.506       | 1.122 | 0.964    | 1097 |
| hidden (256, 128), 300 ep                       | 0.20M  | 0.795     | 0.794 @ 200  | 0.584       | 1.169 | 0.977    | 3392 |
| hidden (512,), 300 ep                           | 0.28M  | 0.795     | 0.795 @ 250  | 0.597       | 1.175 | 0.989    | 3249 |
| hidden (1024, 512), 300 ep                      | 1.59M  | 0.795     | 0.795 @ 299  | 0.556       | 1.149 | 0.981    | 723  |
| hidden (512, 256), 300 ep, {'batch_size': 2048} | 0.53M  | 0.799     | 0.799 @ 299  | 0.612       | 1.175 | 0.989    | 2766 |
| hidden (512, 256), 300 ep, {'batch_size': 128}  | 0.53M  | 0.807     | 0.799 @ 125  | 0.540       | 1.148 | 0.963    | 1144 |
| hidden (512, 256), 100 ep                       | 0.53M  | 0.802     | 0.802 @ 99   | 0.630       | 1.200 | 0.990    | 4633 |
| hidden (1024, 1024, 512), 300 ep                | 3.69M  | 0.822     | 0.822 @ 299  | 0.705       | 1.228 | 0.990    | 8523 |

**L2.ffn** — latent dimension (default architecture):

| m  | val recon (best) | train recon | u     | nearest real | coverage | ESS  |
|----|------------------|-------------|-------|--------------|----------|------|
| 8  | 0.772            | 0.582       | 1.148 | 0.663        | 0.991    | 6063 |
| 16 | 0.725            | 0.492       | 1.096 | 0.611        | 0.990    | 3770 |
| 24 | 0.694            | 0.410       | 1.068 | 0.563        | 0.984    | 2363 |
| 32 | 0.662            | 0.356       | 1.051 | 0.537        | 0.968    | 1247 |
| 48 | 0.614            | 0.287       | 1.038 | 0.501        | 0.935    | 1429 |
| 64 | 0.573            | 0.239       | 1.028 | 0.475        | 0.901    | 1243 |

**L2.ffn** — architecture / training variants at m = 16 (sorted by held-out reconstruction):

| variant                                         | params | val recon | best (epoch) | train recon | u     | coverage | ESS  |
|-------------------------------------------------|--------|-----------|--------------|-------------|-------|----------|------|
| hidden (1024, 1024, 512), 300 ep                | 3.69M  | 0.723     | 0.719 @ 200  | 0.397       | 1.052 | 0.974    | 1845 |
| hidden (512, 512, 256), 300 ep                  | 1.06M  | 0.723     | 0.721 @ 200  | 0.427       | 1.068 | 0.983    | 2903 |
| hidden (512, 256), 300 ep, {'lam_geom': 0.03}   | 0.53M  | 0.722     | 0.722 @ 299  | 0.481       | 1.094 | 0.989    | 3851 |
| hidden (256, 128), 300 ep                       | 0.20M  | 0.723     | 0.722 @ 250  | 0.507       | 1.116 | 0.991    | 4110 |
| default                                         | 0.53M  | 0.725     | 0.725 @ 275  | 0.492       | 1.096 | 0.990    | 3770 |
| hidden (1024, 512), 300 ep                      | 1.59M  | 0.726     | 0.726 @ 275  | 0.486       | 1.093 | 0.989    | 2379 |
| hidden (512, 256), 300 ep, {'lr': 0.001}        | 0.53M  | 0.727     | 0.726 @ 275  | 0.517       | 1.112 | 0.991    | 3351 |
| hidden (512, 256), 300 ep, {'lr': 0.005}        | 0.53M  | 0.728     | 0.728 @ 299  | 0.439       | 1.081 | 0.988    | 3554 |
| hidden (512, 256), 600 ep                       | 0.53M  | 0.736     | 0.728 @ 300  | 0.392       | 1.055 | 0.984    | 2590 |
| hidden (512, 256), 300 ep, {'lam_geom': 0.3}    | 0.53M  | 0.729     | 0.729 @ 299  | 0.496       | 1.103 | 0.989    | 3970 |
| hidden (512,), 300 ep                           | 0.28M  | 0.730     | 0.729 @ 225  | 0.524       | 1.117 | 0.990    | 2823 |
| hidden (512, 256), 300 ep, {'batch_size': 2048} | 0.53M  | 0.731     | 0.731 @ 225  | 0.536       | 1.125 | 0.991    | 3411 |
| hidden (512, 256), 100 ep                       | 0.53M  | 0.735     | 0.735 @ 99   | 0.552       | 1.150 | 0.991    | 4657 |
| hidden (512, 256), 300 ep, {'batch_size': 128}  | 0.53M  | 0.738     | 0.735 @ 150  | 0.468       | 1.099 | 0.986    | 4011 |
| hidden (2048, 1024), 300 ep                     | 5.28M  | 0.751     | 0.751 @ 299  | 0.588       | 1.157 | 0.989    | 7192 |

**L5.ffn** — latent dimension (default architecture):

| m  | val recon (best) | train recon | u     | nearest real | coverage | ESS  |
|----|------------------|-------------|-------|--------------|----------|------|
| 8  | 0.767            | 0.523       | 1.165 | 0.694        | 0.995    | 6780 |
| 16 | 0.704            | 0.411       | 1.114 | 0.620        | 0.991    | 3730 |
| 24 | 0.651            | 0.329       | 1.075 | 0.559        | 0.986    | 2288 |
| 32 | 0.606            | 0.275       | 1.055 | 0.523        | 0.970    | 2170 |
| 48 | 0.542            | 0.211       | 1.038 | 0.491        | 0.925    | 2084 |
| 64 | 0.491            | 0.171       | 1.025 | 0.458        | 0.809    | 1049 |

**L5.ffn** — architecture / training variants at m = 16 (sorted by held-out reconstruction):

| variant                                  | params | val recon | best (epoch) | train recon | u     | coverage | ESS  |
|------------------------------------------|--------|-----------|--------------|-------------|-------|----------|------|
| hidden (512, 512, 256), 300 ep           | 1.06M  | 0.702     | 0.701 @ 225  | 0.366       | 1.085 | 0.992    | 2843 |
| hidden (512, 256), 300 ep, {'lr': 0.005} | 0.53M  | 0.702     | 0.701 @ 250  | 0.376       | 1.095 | 0.992    | 3926 |
| hidden (512, 256), 600 ep                | 0.53M  | 0.702     | 0.701 @ 525  | 0.337       | 1.063 | 0.989    | 1442 |

| variant                                         | params | val recon | best (epoch) | train recon | u     | coverage | ESS  |
|-------------------------------------------------|--------|-----------|--------------|-------------|-------|----------|------|
| hidden (512, 256), 300 ep, {'lam_geom': 0.03}   | 0.53M  | 0.704     | 0.704 @ 299  | 0.404       | 1.114 | 0.993    | 4860 |
| default                                         | 0.53M  | 0.704     | 0.704 @ 250  | 0.411       | 1.114 | 0.991    | 3730 |
| hidden (512, 256), 300 ep, {'lam_geom': 0.3}    | 0.53M  | 0.706     | 0.705 @ 250  | 0.416       | 1.113 | 0.991    | 4357 |
| hidden (1024, 512), 300 ep                      | 1.59M  | 0.706     | 0.706 @ 275  | 0.410       | 1.115 | 0.992    | 3417 |
| hidden (512, 256), 300 ep, {'batch_size': 128}  | 0.53M  | 0.708     | 0.708 @ 299  | 0.391       | 1.104 | 0.992    | 3322 |
| hidden (512, 256), 300 ep, {'lr': 0.001}        | 0.53M  | 0.708     | 0.708 @ 275  | 0.432       | 1.124 | 0.992    | 3882 |
| hidden (256, 128), 300 ep                       | 0.20M  | 0.709     | 0.709 @ 299  | 0.425       | 1.131 | 0.991    | 4370 |
| hidden (512, ), 300 ep                          | 0.28M  | 0.716     | 0.716 @ 250  | 0.438       | 1.126 | 0.993    | 4706 |
| hidden (512, 256), 300 ep, {'batch_size': 2048} | 0.53M  | 0.718     | 0.718 @ 275  | 0.454       | 1.139 | 0.991    | 4302 |
| hidden (512, 256), 100 ep                       | 0.53M  | 0.721     | 0.721 @ 99   | 0.465       | 1.155 | 0.992    | 5954 |
| hidden (2048, 1024), 300 ep                     | 5.28M  | 0.725     | 0.725 @ 275  | 0.474       | 1.150 | 0.991    | 6530 |
| hidden (1024, 1024, 512), 300 ep                | 3.69M  | 0.825     | 0.825 @ 299  | 0.648       | 1.223 | 0.981    | 5117 |

## 8. How the samples look to the model (5M\_downstream, SimpleStories/SimpleStories-5M)

Embedding samples are placed at the last position of the prefix and the *full* model is run. (a) Next-token distributions: Jensen-Shannon divergence to the distribution of the sample's nearest real token, second-nearest, and a random token; entropy; top-1 agreement. (b) Layer-by-layer trajectory of the images against every location's own manifold and bands, with perp = distance from the segment joining the images of the two nearest real anchors (in spacings) and the fraction of images whose projection falls between those two anchor images.

| set                  | JS to nearest token | JS to 2nd nearest | JS to random token | entropy (real: 2.86) | top-1 = nearest | top-1 = random |
|----------------------|---------------------|-------------------|--------------------|----------------------|-----------------|----------------|
| held-out real tokens | 0.457               | 0.515             | 0.606              | 2.800                | 0.193           | 0.038          |
| samples $\alpha=0.3$ | 0.212               | 0.291             | 0.565              | 3.380                | 0.316           | 0.064          |
| samples $\alpha=0.7$ | 0.274               | 0.327             | 0.554              | 3.272                | 0.260           | 0.067          |
| samples $\alpha=1.5$ | 0.368               | 0.396             | 0.542              | 2.858                | 0.216           | 0.071          |
| real + 1x noise      | 0.508               | 0.522             | 0.575              | 3.883                | 0.076           | 0.043          |

Examples ( $\alpha$  = first sweep value): the two nearest real tokens of a sample, the sample's top-3 next-token predictions, and the nearest token's top-3.

| nearest | 2nd nearest | sample predicts | nearest token predicts |
|---------|-------------|-----------------|------------------------|
| himself | herself     | . , and         | . , and                |
| guiding | bringing    | them him away   | star light place       |
| ##elt   | pizz        | ##a ##er ##ous  | ##a ##pt ##er          |
| jose    | leo         | who . ,         | and . ,                |
| cool    | cold        | and , -         | , ##er dinosaur        |
| clouds  | snowflakes  | . , that        | that of .              |

kernel score u of the images, per location:

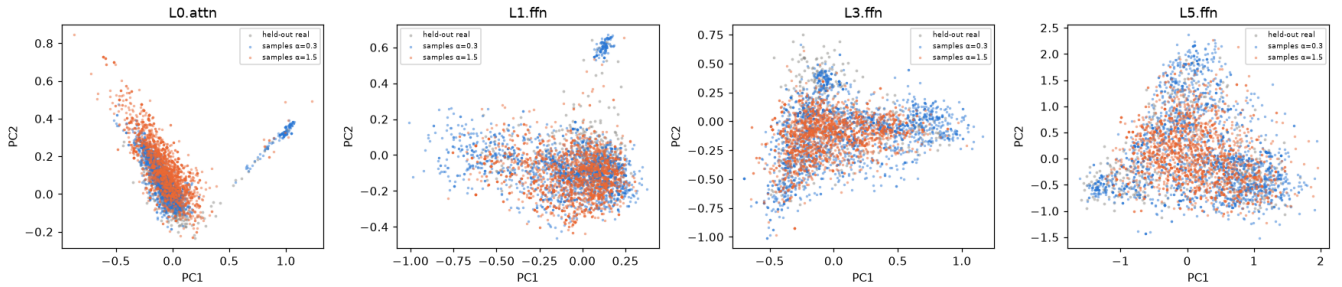
| location | samples $\alpha=0.3$ | samples $\alpha=0.7$ | samples $\alpha=1.5$ | held-out real | real + 1x noise | band: real + 1x noise |
|----------|----------------------|----------------------|----------------------|---------------|-----------------|-----------------------|
| embed    | 1.149                | 1.088                | 0.758                | 0.995         | 0.707           | 0.707                 |
| L0.attn  | 1.144                | 1.084                | 0.768                | 0.994         | 0.707           | 0.721                 |
| L0.ffn   | 1.035                | 1.008                | 0.795                | 1.001         | 0.781           | 0.695                 |
| L1.attn  | 1.021                | 0.997                | 0.796                | 1.000         | 0.783           | 0.694                 |
| L1.ffn   | 1.000                | 0.984                | 0.842                | 1.000         | 0.847           | 0.690                 |
| L2.attn  | 0.998                | 0.983                | 0.848                | 1.000         | 0.847           | 0.690                 |
| L2.ffn   | 0.996                | 0.989                | 0.903                | 0.998         | 0.891           | 0.691                 |
| L3.attn  | 0.996                | 0.988                | 0.903                | 0.999         | 0.902           | 0.692                 |
| L3.ffn   | 0.997                | 0.995                | 0.951                | 0.999         | 0.942           | 0.689                 |
| L4.attn  | 0.998                | 0.994                | 0.943                | 0.998         | 0.946           | 0.685                 |
| L4.ffn   | 0.999                | 0.998                | 0.968                | 0.997         | 0.971           | 0.686                 |
| L5.attn  | 0.999                | 0.997                | 0.965                | 0.998         | 0.972           | 0.686                 |
| L5.ffn   | 0.995                | 0.993                | 0.974                | 0.998         | 0.995           | 0.697                 |

nearest real / spacing of the images, per location:

| location | samples $\alpha=0.3$ | samples $\alpha=0.7$ | samples $\alpha=1.5$ | held-out real | real + 1x noise | band: real + 1x noise |
|----------|----------------------|----------------------|----------------------|---------------|-----------------|-----------------------|
| embed    | 0.642                | 0.737                | 1.150                | 0.948         | 1.368           | 1.369                 |
| L0.attn  | 0.655                | 0.748                | 1.162                | 0.965         | 1.388           | 1.379                 |
| L0.ffn   | 0.741                | 0.808                | 1.134                | 0.902         | 1.249           | 1.320                 |
| L1.attn  | 0.759                | 0.819                | 1.135                | 0.899         | 1.247           | 1.314                 |
| L1.ffn   | 0.796                | 0.849                | 1.113                | 0.911         | 1.203           | 1.308                 |
| L2.attn  | 0.799                | 0.851                | 1.116                | 0.910         | 1.205           | 1.307                 |
| L2.ffn   | 0.813                | 0.854                | 1.083                | 0.895         | 1.162           | 1.309                 |
| L3.attn  | 0.809                | 0.858                | 1.084                | 0.897         | 1.152           | 1.303                 |
| L3.ffn   | 0.814                | 0.863                | 1.056                | 0.904         | 1.103           | 1.301                 |
| L4.attn  | 0.817                | 0.873                | 1.065                | 0.899         | 1.110           | 1.309                 |
| L4.ffn   | 0.822                | 0.881                | 1.050                | 0.913         | 1.095           | 1.307                 |
| L5.attn  | 0.825                | 0.887                | 1.062                | 0.919         | 1.087           | 1.314                 |
| L5.ffn   | 0.858                | 0.911                | 1.053                | 0.920         | 1.051           | 1.325                 |

perp. distance from the anchor-image segment / spacing of the images, per location:

| location | samples $\alpha=0.3$ | samples $\alpha=0.7$ | samples $\alpha=1.5$ | held-out real | real + 1x noise |
|----------|----------------------|----------------------|----------------------|---------------|-----------------|
| embed    | 0.517                | 0.620                | 1.063                | 0.826         | 1.277           |
| L0.attn  | 0.532                | 0.632                | 1.073                | 0.840         | 1.300           |
| L0.ffn   | 0.653                | 0.712                | 1.056                | 0.808         | 1.186           |
| L1.attn  | 0.676                | 0.731                | 1.061                | 0.813         | 1.186           |
| L1.ffn   | 0.718                | 0.764                | 1.046                | 0.825         | 1.165           |
| L2.attn  | 0.722                | 0.767                | 1.050                | 0.829         | 1.164           |
| L2.ffn   | 0.740                | 0.784                | 1.022                | 0.851         | 1.150           |
| L3.attn  | 0.740                | 0.787                | 1.027                | 0.848         | 1.145           |
| L3.ffn   | 0.751                | 0.794                | 1.005                | 0.862         | 1.122           |
| L4.attn  | 0.756                | 0.799                | 1.012                | 0.857         | 1.123           |
| L4.ffn   | 0.763                | 0.818                | 1.018                | 0.907         | 1.121           |
| L5.attn  | 0.768                | 0.829                | 1.037                | 0.907         | 1.146           |
| L5.ffn   | 0.810                | 0.870                | 1.054                | 0.950         | 1.140           |



PCA of the real states (grey) and of the images of samples at the smallest (blue) and largest (orange)  $\alpha$ , at four locations.